

Lab 04 Reflection Journal

Exploratory Data Analysis: Titanic Dataset

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What the lab was about

This lab introduced Exploratory Data Analysis (EDA) using the Titanic passenger dataset. We used pandas to inspect and summarize the data, and Seaborn/Matplotlib to visualize how passenger characteristics relate to survival. The goal was to understand the data before thinking about any model.

Key findings

- **The dataset is imbalanced.** Only about 38% of passengers survived. A model that always predicted "died" would already be about 62% accurate, which matters when evaluating models later.
- **Gender and class were the strongest patterns.** Females survived at a much higher rate than males, and 1st class passengers survived far more often than 3rd class passengers.
- **Age mattered mainly for young children.** The survivor histogram showed a clear spike for small children, supporting the "women and children first" idea.
- **Fare reflects wealth.** Survivors had a median fare of about \$26, compared with about \$10.50 for those who died. Fare is closely tied to passenger class.
- **Port of embarkation was misleading at first.** Cherbourg passengers survived at about 55%, compared with about 34% for Southampton. But about half of Cherbourg passengers were 1st class, while Queenstown was over 90% 3rd class. The port itself likely did not affect survival; it was standing in for class.

What I learned

The most important lesson for me was that a relationship in the data does not always mean one variable causes the other. The embarkation result looked meaningful until I compared it with passenger class. EDA is not just making plots; it is asking follow-up questions about why a pattern shows up.

I also learned why plots matter in addition to summary statistics. The mean fare (about \$32) made fares look moderate, but the boxplot showed that half of passengers paid under about \$15 while a few paid over \$500. I also noticed data quality issues that would need handling before modeling: *Age* and *Cabin* have many missing values, and *Embarked* is missing 2 values.

Challenges

One challenge was that raw counts were misleading in Experiment 1. Southampton had by far the most survivors in total simply because most passengers boarded there, so I had to calculate the survival *rate* for each port to compare them fairly. Another challenge was in Experiment 2: the extreme fares (up to about \$512) squashed the violin plot so the distribution was hard to read. I solved this by showing the full boxplot next to a violin plot limited to fares of \$300 or less.

Next steps and questions

- How should the missing *Age* values be handled: fill them with the overall median, or with a median by class or title?
- Would combining features (for example, class and gender together) give a clearer picture than looking at each one separately?
- How will the class imbalance affect model training in later labs?